

Collaborative Ocular Melanoma Study randomized trial of I-125 brachytherapy

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Background Although findings from observational studies published in the 1970s and early 1980s suggested that length of remaining life after a diagnosis of uveal melanoma was similar following enucleation (removal of the eye) and local eye-conserving radiotherapy, the majority of ophthalmologists in the United States were not convinced that “saving the eye” would not compromise survival.

Purpose The Collaborative Ocular Melanoma Study (COMS) was designed to compare enucleation and radiotherapy among similar patients with respect to survival outcomes.

Methods A multicenter randomized trial of primary treatment with iodine-125 brachytherapy versus enucleation (i.e., a comparative effectiveness trial) was conducted to provide the evidence required to answer the concerns of ophthalmologists and their patients. Eligibility criteria adopted for the trial were intended to apply to the majority of patients diagnosed with choroidal melanoma who would be suitable candidates for either form of primary treatment.

Results Accrual to this COMS trial began in 1986 and ended in 1998. Participating patients were followed for 5 to 15 years, depending upon date of enrollment, before all clinical follow-up ended in 2003. No difference in survival outcomes and little difference in quality-of-life outcomes were observed between patients in the brachytherapy arm and those in the enucleation arm. Five-year survival was substantially better than anticipated based on a review of the literature when the trial was designed.

Limitations The choice of brachytherapy for delivery of radiation to the tumor had important advantages but also imposed restrictions regarding eligibility.

Conclusions The multidisciplinary COMS Group not only successfully conducted a randomized trial that answered the primary study questions but also has contributed to clinical and epidemiologic knowledge of choroidal melanoma through numerous publications. As a consequence of the COMS, a standard approach to I-125 brachytherapy for treatment of choroidal melanoma became widely available to affected patients. *Clinical Trials* 2011; 8: 661–673. <http://ctj.sagepub.com>

Introduction

Despite its low incidence (six to eight cases per million per year in the USA), melanoma of the uveal tract is the most common primary eye cancer in adults. The most frequent site of metastasis is the liver

but metastatic dissemination to almost every organ and tissue in the body has been reported. Although bilateral cases of uveal melanoma have been reported, this cancer is almost always unilateral.

To date, the Collaborative Ocular Melanoma Study (COMS) Group has conducted the only

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randomized clinical trials of treatment for a life-threatening ocular condition in which mortality has been the primary (design) outcome. The COMS trials were designed and conducted amidst considerable controversy. The tension in the ophthalmological community was due to the possibly incompatible goals of prolonging life versus retaining vision. The COMS design included two randomized trials of radiotherapy and a non-randomized prospective study [1]. In the COMS trial of primary interest, randomization was between enucleation (removal of the eye) and local tumor control via radiotherapy with the goal of retaining the eye and, possibly, retaining some useful vision in it. The other COMS trial was for patients whose tumors were too large or in a location that made local control via radiotherapy impractical; randomization was between enucleation alone and enucleation with pre-operative adjuvant radiotherapy.

Background

In the late 1970s and early 1980s, most US ophthalmologists who diagnosed uveal melanoma in a patient confronted a difficult decision, i.e., whether to recommend removal of the eye containing the tumor, in accord with a century of ophthalmologic practice, or to refer the patient to one of a small number of ophthalmologists who advocated eye-conserving radiotherapy for many such tumors [2–5]. Because most ophthalmologists see few, if any, patients with ocular melanoma in a lifetime of clinical practice, many elected to refer patients with a suspicious ocular mass to a few ophthalmologists with training in ocular oncology who had developed considerable experience with diagnosing and treating ocular tumors. Several ophthalmologists in this small group had devoted a significant portion of their careers to the development of radiotherapy methods for treating ocular tumors and were convinced of their effectiveness. The dilemma for the ophthalmologist and patient was exacerbated by published analysis of a large series of patients who had undergone enucleation that showed that the period with the highest incidence of death was 1–2 years following enucleation [6]. Some readers concluded that enucleation was responsible for disseminating tumor cells during surgery that led to metastasis and death. Also, the ophthalmologist had to consider the possibility of removing an eye that harbored not melanoma but a benign mass; despite decreasing rates of misdiagnosis, unnecessary enucleation remained a concern. Thus, the ophthalmologist had to choose between two options: remove the affected eye and possibly hasten the

patient's death (or discover a misdiagnosis) or refer the patient for radiotherapy and possibly hasten his or her death should radiotherapy prove to be less effective than enucleation to control the tumor and avoid metastasis [7].

As survival data from uncontrolled series of patients treated with eye-conserving radiotherapy accumulated and comparisons were made to historical series of patients whose tumors were managed by enucleation [8,9], the available evidence suggested that enucleation and radiotherapy were equally effective (or equally ineffective) with respect to prolonging life. However, because of the retrospective nature of the comparisons and differences in the characteristics of patients and tumors selected for enucleation and radiotherapy, most ophthalmologists in the USA remained uncertain regarding the better course for their patients. Furthermore, enucleation could be performed by the patient's local ophthalmologist, in contrast to radiotherapy. The largest US clinical practices specializing in radiotherapy for uveal melanoma were located on the Atlantic and Pacific coasts of the USA; thus, most patients who elected eye-conserving radiotherapy incurred significant expense associated with travel for treatment and post-treatment follow-up by the physicians at those centers.

Initiation and design

Initiation

During the period 1978–1983, several proposals to conduct single-center and multi-center studies of treatment of ocular melanoma were submitted to the National Institutes of Health. Although each of these proposals had merit, none was funded because either the National Eye Institute, the National Advisory Eye Council, or both concluded that each proposal addressed a limited aspect of the totality of issues regarding survival, local tumor control, side effects of treatment, and quality of life.

A plea for a randomized trial was made in June 1980 by Stuart L. Fine, MD, and Argye I. Hillis, PhD, (Wilmer Eye Institute, the Johns Hopkins University School of Medicine, Baltimore) during the annual Wilmer Retinal Vascular Center meeting. Subsequently, Carl Kupfer, MD, Director of the National Eye Institute (National Institutes of Health, US Department of Health and Human Services, Bethesda, MD) and an advocate of randomized clinical trials, appointed Thomas Duane, MD, (Wills Eye Hospital, Thomas Jefferson University, Philadelphia) to chair an *ad hoc*

committee to address the problem of ocular melanoma therapy. This committee, which included representatives from the National Eye Institute and National Cancer Institute as well as consultants from ophthalmology, radiation oncology, radiation physics, and biostatistics, recommended a comprehensive study of ocular melanoma, including a collaborative clinical trial to compare radiotherapy with enucleation in suitable patients. However, the ophthalmologic community was slow to act.

During a series of meetings and discussions that Dr Fine initiated with retinal specialists from the USA and Europe in 1983 and 1984, a consensus gradually emerged that a scientifically rigorous prospective study was needed of comparable cases of uveal melanoma (primary melanoma arising in the choroid or ciliary body), some of whom were treated with enucleation and others by eye-conserving radiotherapy. However, the appropriate design of such a study was debated in multiple venues. In February 1984, Dr Kupfer asked Bradley R. Straatsma, MD, (Jules Stein Eye Institute, University of California Los Angeles) to chair an *ad hoc* committee to reassess the problem of treatment of uveal melanoma, with emphasis on the feasibility of a randomized clinical trial. Subsequently, Drs Straatsma and Fine agreed to collaborate to achieve their shared goals. In March 1984, the committee chaired by Dr Straatsma issued a report that echoed the conclusions of the committee chaired by Dr Duane and called for a collaborative trial of management of uveal melanoma.

Several meetings were held later in 1984 to design a study, with many ophthalmologists and biostatisticians advocating a randomized clinical trial but with ophthalmologists who were proponents of radiotherapy arguing for an observational study. Many of those who favored a randomized trial were retinal specialists who had participated or were participating in randomized trials of treatment of other eye conditions (primarily diabetic retinopathy, branch vein occlusion, and choroidal neovascularization) and were convinced of the value of such a study design.

The debate between these two factions reached a climax at a meeting held in Bethesda, MD, in December 1984 under the auspices of the National Eye Institute. At that time, the two methods of eye-conserving radiotherapy for which the largest experience was available in the USA were charged particles (either protons or helium ions), used for treating uveal melanoma primarily at two sites, and brachytherapy via a radioactive plaque sutured to the sclera (outside of the eye) overlying the base of the melanoma, also used primarily at two sites. Several designs for one or more randomized trials that included both methods

of delivering radiotherapy were presented by participants who advocated such designs. Designs offered for consideration included, among others, a three-arm trial in which eligible patients would be assigned to enucleation, radiotherapy via an episcleral plaque, or charged particle radiation in a 1:1:1 ratio; a three-arm trial with patients assigned equally to enucleation or radiotherapy but with patients assigned to radiotherapy further assigned at random to charged particles or radioactive plaque so that the resulting ratio would be 2:1:1; two parallel trials, i.e., enucleation versus radioactive plaque and enucleation versus radiotherapy with charged particles, so that the patient and ophthalmologist could elect the trial in which the patient would enroll. The logistics of conducting one or more randomized trials were discussed at length and included potential arrangements to transport patients assigned to radiotherapy to one of the major ophthalmic radiotherapy centers.

Ultimately, efforts to construct a study design in which all retinal specialists, including radiotherapy proponents, could agree to participate by referral of patients ended in failure. In fact, some radiotherapy proponents charged that randomization was unethical because they believed that eye-conserving therapy was better for most patients, some of whom would be assigned unjustifiably, in their opinion, to enucleation in a randomized trial to compare the two treatment approaches. Furthermore, as a group, they refused to participate by enrolling or referring eligible patients and also refused to provide radiotherapy for patients so assigned who enrolled in a randomized trial elsewhere. After taking this position, the radiotherapy proponents elected to withdraw from further consideration of a randomized trial. However, because a majority of ophthalmologists and Dr Kupfer believed that a randomized trial was essential to compare radiotherapy to enucleation with respect to survival, planning for such a study moved ahead under the leadership of a dedicated team of researchers headed primarily by Drs Straatsma and Fine.

The withdrawal of the radiotherapy proponents from deliberations resulted in withdrawal of the ophthalmologist who had expected to be the principal investigator of the coordinating center for one or more randomized trials because of continuing collaboration with a strong opponent of a randomized trial. Consequently, two experienced clinical trials methodologists from the Wilmer Eye Institute (Barbara S. Hawkins, PhD, and Maureen G. Maguire, PhD) were recruited to participate in the planning process with the expectation that the study coordinating center would be located at the Johns Hopkins University. Their activities included review of methods and

analyses used in published observational studies. Together with Andrew P. Schachat, MD, (Wilmer Eye Institute), Drs Fine, Maguire, and Hawkins met with Dr Kupfer on February 20, 1985. After discussing a number of issues, the Wilmer investigators agreed to work with interested retinal specialists to prepare a detailed protocol for a multi-center clinical trial of enucleation versus radiotherapy, with mortality as the primary outcome. Although Dr Kupfer requested that funding proposals be submitted by April 1, 1985, to take advantage of a special allocation in the National Eye Institute's 1985 budget to fund important new clinical trials, Dr Hawkins insisted that more time would be required; thus, early May 1985 became the target date for submission.

Design

Following the withdrawal of the major radiotherapy advocates from discussions, design decisions focused on the number of clinical trials to be conducted, the method(s) of delivering radiotherapy and the number of treatment arms in each trial, and the effect size that would lead to a preference for one treatment over another if a difference in 5-year mortality rates of that magnitude were observed. Initially, three randomized trials were considered, one each for large, intermediate, and small uveal melanoma. Consensus rapidly was achieved regarding the design and treatment arms for randomized trials for large and 'medium-sized' tumors. However, the risk of misdiagnosis was believed to be high in small tumors; many possibly were large nevi. Therefore, neither enucleation nor eye-conserving radiotherapy was deemed acceptable for these small, possibly benign, lesions because of the potentially unfavorable imbalance between risks, i.e., loss of the eye in the case of enucleation or likely vision-threatening side effects of radiotherapy, and survival benefit. Consequently, design of a randomized trial for small uveal melanoma was tabled.

The primary trial of interest was designed to compare enucleation with eye-conserving radiotherapy for a majority of patients with uveal melanoma, defined to be of 'medium size,' who would be suitable candidates for either treatment. Brachytherapy was selected for the radiotherapy arm in this trial because of the small number of facilities equipped to deliver charged particles and refusal of investigators at those facilities to participate in a randomized trial. The second randomized trial was designed to evaluate radiotherapy as an adjunct to enucleation with the goal of improving the survival rate among patients

with large uveal melanoma for whom most ophthalmologists would recommend enucleation. (The radiation dose required for adjunctive treatment of large uveal melanoma had been demonstrated to have unacceptable local adverse effects so that removal of the eye was required.) Investigators at the Radiological Physics Center (Houston, TX), led by William F. Hanson, PhD, who were providing radiation physics expertise to a number of radiotherapy trials in cancer, and a radiation oncologist experienced in clinical trials, John D. Earle, MD, (Mayo Foundation, Rochester, MN) joined the study organizers and, in collaboration with ophthalmologists, oncologists, and radiation physicists, developed a standard brachytherapy protocol that could be implemented at many institutions and monitored centrally [10].

A review and meta-analysis of the published mortality rates for uveal melanoma undertaken by faculty members at the Wilmer Eye Institute suggested that the 5-year all-cause mortality rate among patients treated with enucleation was 30% when the tumor was medium-sized and 50% when it was large [11]. With two treatment arms in each of the clinical trials and the goal of excellent statistical power ($\geq 90\%$) to detect or rule out a relative treatment difference of 25% with high probability ($\geq 99\%$, i.e., $\alpha \leq 0.01$), sample sizes of 2400 and 1000, respectively, were projected for a trial of brachytherapy versus enucleation for medium-sized tumors and a trial of pre-enucleation external beam radiation versus enucleation alone for large tumors [1].

During the protocol development period, the members of the COMS planning group agreed to focus the clinical trials on *choroidal* melanoma because of typically earlier diagnosis but worse prognosis for more anteriorly located uveal melanoma, i.e., those arising in the iris or the ciliary body. Because uveal melanoma is a relatively rare primary cancer, it was recognized that a large number of participating clinical centers would be required and that it would be necessary to modify referral patterns established during the previous decade, i.e., referral of a significant proportion of patients with suspected uveal melanoma to ophthalmologists participating in the COMS rather than to the major radiotherapy centers. Furthermore, with 5-year mortality as the projected outcome of primary interest, patients would have to be followed for at least 5 years or until death, requiring a long-term study to recruit and follow a sufficient number of patients.

An important design issue was the amount and type of data to be collected. There was consensus that it was important that the identity of patients with choroidal melanoma screened for the COMS be confidential; thus, no personal identifying

information for any patient screened or enrolled would be collected centrally. With mortality as the primary outcome and a large sample size, the planning group recognized that it would be feasible to conduct the proposed clinical trials as 'large, simple trials' in which eligible consenting patients would be assigned centrally to treatment via randomization and deaths would be identified periodically from the National Death Index and other public databases or reported by personnel at the enrolling centers as they became aware of them. With such a design, data collection would be limited to that required to ascertain eligibility and to provide baseline descriptors of the enrollees and their tumors in addition to date of death. However, arguments were made for a structured schedule of follow-up clinical examinations and more extensive data collection to document protocol adherence and clinical outcomes, such as rate of visual acuity loss, patterns of tumor regression, and radiation retinopathy development and progression, in eyes treated with brachytherapy. Furthermore, the cause of death was considered essential information because primary treatment of the tumor was expected to affect only deaths resulting from melanoma metastasis but not those from other causes. Because the expectation was that most patients with choroidal melanoma identified at study centers during the period of accrual would be referred to the COMS, thus limiting the ability of participating investigators to conduct research into other types of treatment and to investigate ancillary issues, agreement was reached to collect substantially more data from COMS patients than the minimum required for the primary comparison of treatments.

Critics of the study expressed doubt that a sufficient number of patients could be enrolled in the randomized trials to provide scientifically valid comparisons of treatment arms with respect to mortality outcomes. They also questioned whether protocols for diagnosis and treatment of choroidal melanoma could be designed and implemented in a standard fashion across multiple institutions. These issues also were of concern to personnel at the National Eye Institute (the expected sponsor) and to the investigators who were designing the COMS.

To permit the COMS Steering Committee and Data and Safety Monitoring Committee to monitor the feasibility of enrolling a sufficient number of patients in the COMS clinical trials within a reasonable time period, the COMS design included reporting of each patient diagnosed with choroidal melanoma at a participating COMS center during the accrual period. Detailed baseline data were collected and reported for enrolling patients. A few demographic characteristics in

addition to tumor dimensions and location were reported for patients judged by the local COMS ophthalmologist to be ineligible for a COMS trial and for patients judged to be eligible but who did not enroll. Reasons for ineligibility or not enrolling were reported along with the treatment selected for each of these latter patients [12,13]. Consideration was given to requiring that an 'informed refusal form,' complementary to the informed consent form to be signed by enrolling patients, be signed by patients who were eligible for the COMS but who did not enroll. At the time of their review of the study protocol, the Data and Safety Monitoring Committee judged that such a form would be coercive; thus, this provision was deleted. Also, mortality follow-up of patients who were eligible for one of the two randomized trials but did not enroll was proposed to permit future comparison of mortality rates between the group of eligible non-enrollers and enrollees. Objections to this aspect of the study design during preliminary review by the lead institutional review board, together with concerns about the feasibility and cost, resulted in deletion of this design feature.

In addition to a common protocol followed at all clinical centers, standard data collection forms, and concurrent central data management and monitoring, quality assurance mechanisms implemented in the COMS included computer-assisted central review of eligibility by personnel at the COMS Coordinating Center before a random treatment assignment was made; central review of echograms and photographs taken of the tumor at specialized resource centers; central review of the histopathology of enucleated eyes; central verification of radiation dosimetry calculations; and classification of decedents by presence or absence of melanoma metastasis at death by an independent Mortality Coding Committee. Independent study oversight was provided by a Data and Safety Monitoring Committee (Matthew D. Davis, MD, Chair) appointed by the sponsor.

Conduct phase

In September 1985, the National Eye Institute awarded funding for four proposed COMS resource centers and 19 clinical centers to permit investigators at these centers to participate in refinement of the COMS protocol and procedures. Additional clinical centers and two more resource centers were funded in 1986. A chronology of significant events in the history of the COMS is provided in Table 1.

Table 1 Chronology of significant events in the COMS randomized trial of enucleation vs iodine-125 brachytherapy 1980–2008

Date	Event
1980–1984	Discussions in ophthalmologic community of need for scientifically rigorous comparison of mortality rates among patients with uveal melanoma treated with enucleation vs eye-conserving radiotherapy
May–June, 1985	Submission of funding proposals for COMS, including draft of COMS Manual of Procedures and applications from six resource centers and 30 clinical centers, to National Eye Institute
September 19, 1985	Funding awards to 4 resource centers and 19 clinical centers
July 1, 1986	Funding awards to two remaining resource centers and additional clinical centers
July 15–16, 1986	Protocol review meeting for COMS personnel at all participating centers
August 27, 1986	Approval of COMS protocol by Data and Safety Monitoring Committee
November 5, 1986	First site visit to a COMS clinical center by leadership team
January 22, 1987	Enrollment of first patient in COMS trial of enucleation vs iodine-125 brachytherapy
October 1, 1988	Funding awards to additional clinical centers
September 19, 1989	First review of mortality data of treatment arm by Data and Safety Monitoring Committee
January 1990	Approval of 5-year continuation of funding by National Advisory Eye Council
June 1994	Approval of 5-year continuation of funding by National Advisory Eye Council
August 1994	Funding award for COMS Quality-of-Life Study
July 31, 1998	Completion of patient enrollment in the COMS randomized trial of enucleation vs iodine-125 brachytherapy and COMS Quality-of-Life Study
February 10, 2000	Approval of final 5-year funding plan by National Advisory Eye Council
July 2001	Publication of interim mortality findings from COMS trial of enucleation vs iodine-125 brachytherapy
July 31, 2003	End of clinical data collection
September 28, 2003	Last clinic monitoring site visit to a COMS clinical center
September 29–30, 2003	Final meeting of COMS personnel from all participating centers
November 5, 2003	Final in-person meeting of Data and Safety Monitoring Committee
January 31, 2004	End of funding for clinical center operations
July 31, 2004	End of funding for resource centers other than Coordinating Center
December 31, 2005	End of funding for all COMS activities
December 2006	Publication of end-of-study mortality findings from COMS randomized trial of enucleation vs iodine-125 brachytherapy
August 2008	Deposit of COMS database, reduced dataset, study documents, set of COMS publications to date, and microfilm copies of data collection forms for each COMS patient in the Alan Mason Chesney Medical Archives of Johns Hopkins University

Preparations for patient screening and enrollment

During the first year of funding, substantial effort was devoted to preparation of educational materials for patients with choroidal melanoma and their physicians. Educational materials eventually included those directed at potential patient participants, i.e., illustrated booklets and videotapes that described methods for diagnosing and treating choroidal melanoma and the design and rationale for the COMS clinical trials, and physician-oriented brochures and exhibits for display at professional meetings.

A protocol review and training meeting for all COMS personnel was held in Baltimore in July 1986 and was attended by more than 300 ophthalmologists, oncologists, radiation physicists, echographers, photographers, clinic coordinators, ophthalmic technicians, and resource center personnel. (Figure 1 shows members of the COMS Executive Committee and guests who met just prior to the larger meeting.) This meeting was

important for informing personnel from participating centers about the rationale and design of the COMS and for assuring that the procedures developed by the planning group could be implemented at all clinical centers. Such training was the first step in COMS certification to perform individual study roles [14].

The Data and Safety Monitoring Committee reviewed the design of the COMS at their first meeting in August 1986 before patient screening and enrollment began. Site visits to each participating clinical center were conducted after local approval of the study protocol as the individual center met COMS requirements. Ultimately, the COMS trials were conducted at more than 50 institutions in the USA and Canada and supported by six resource centers.

Patient accrual and treatment

The first patient was enrolled in the trial of brachytherapy versus enucleation in January



Figure 1 Participants of the July 15, 1986, meeting of the COMS Executive Committee held immediately prior to the protocol review meeting for all COMS investigators and personnel.

First row (left to right): Bradley R Straatsma, MD, COMS Co-Chair; Richard Margolese, MD, Guest; Stuart L Fine, MD, COMS Chair; Carl Kupfer, MD, Director of the National Eye Institute; and Barbara S Hawkins, PhD, Principal Investigator of COMS Coordinating Center.

Second row (left to right): Peter Fitzpatrick, MBBS; Daniel M Albert, MD, Principal Investigator of COMS Pathology Center; Jane Ebert, Assistant to COMS Chair; Marguerite Alexander, RN; Sandra F Byrne, Principal Investigator of COMS Echography Center; Paul Carbone, MD; Jack A McLaughlin, PhD, NEI Program Staff Liaison; Fran Goff, NEI Staff Guest; and Marie Diener-West, PhD, COMS Statistician and Co-Investigator of COMS Coordinating Center.

Third row (left to right): John D Earle, MD, COMS Co-Chair; William F Hanson, PhD, Principal Investigator of Radiologic Physics Center; Thomas A Weingeist, MD, PhD, Principal Investigator of COMS Photograph Reading Center; Dennis M Robertson, MD; Jan A Markowitz, PhD, COMS Protocol Monitor; Robert Harrison, COT; and Andrew P Schachat, MD, Assistant COMS Chair.

Back row (left to right): Thomas A Rice, MD and Travis A Meredith, MD.

1987, following enrollment of the first patient in the companion trial in November 1986. Accrual to the brachytherapy trial concluded in July 1998, as recommended by the Data and Safety Monitoring Committee. (Accrual to the companion trial ended in December 1994.) The COMS investigators successfully changed referral patterns [15] in order to accrue a sufficient number of patients. Other measures taken to optimize accrual included adding clinical centers during the course of the trial [16] and 'marketing' the study to physicians and patients through a website designed to complement other informational sources.

Periodic review of the information reported for all patients diagnosed with choroidal melanoma permitted the Steering Committee to evaluate the

effect of individual eligibility and exclusion criteria on patient accrual and to reconsider aspects of the protocol that could be modified to increase participation. A major goal of the brachytherapy trial was to include a sample representative of the majority of patients diagnosed with choroidal melanoma so that findings would have broad applicability. In 1991, it became apparent that many otherwise eligible patients with tumors smaller than the lower height criterion (3 mm) or larger than the upper height criterion (8 mm) originally adopted for this trial were being treated with brachytherapy and enucleation outside the COMS in nearly equal numbers. After discussing the ramifications of expanding the tumor size criteria for the brachytherapy trial with the Data and Safety Monitoring

Committee and the study investigators, the Steering Committee modified the criteria to be more inclusive.

When the COMS radiotherapy protocol was developed, many assumptions were required to make the dose calculations for iodine-125 brachytherapy using the gold alloy episcleral shields. Research by radiation physicists at the Radiological Physics Center and elsewhere, under the auspices of the American Association of Physicists in Medicine, continued while the COMS brachytherapy trial was underway [17]. This work resulted in relabeling of the target dose to the tumor apex prescribed by the COMS protocol, i.e., as 85 Gy based on better data rather than 100 Gy as originally calculated, without any change in the actual radiation dose delivered.

Data analysis, monitoring, and publication

Data quality

The result of the quality assurance methods adopted by the COMS Group was a very high quality study in which few ineligible patients enrolled, almost all patients received the assigned treatment shortly after enrollment, follow-up for mortality and clinical outcomes was excellent, and adherence to the radiotherapy protocols was exceptionally good based on assessments at the Radiological Physics Center and by Dr Earle [18–21] and comparison with other trials of radiotherapy monitored by them.

Data analysis, data monitoring, and outcomes

The primary method of analysis of the mortality outcomes in the COMS was the product-limit method with rates by treatment arm compared graphically and with the log-rank test. The Data and Safety Monitoring Committee decided at their first meeting that they would not look at mortality data by treatment arm during the first 5 years of the trials to avoid the possibility of reaching a premature conclusion based on a small fraction of the target sample size and short follow-up. However, they agreed to do so in 1990 when continuation funding was requested from the National Eye Institute. Because of the expected length of the study, the Committee agreed, at the request of the investigators, to release interim mortality data from each trial after the 5-year

outcome was known for 75% of the patients already enrolled.

In 1997, the COMS Data and Safety Monitoring Committee approved publication of interim mortality findings from the trial of pre-enucleation radiation for large choroidal melanoma [20]. This decision was important to sustain the commitment of the study investigators to continued accrual to the brachytherapy trial. In 2001, that Committee approved publication of interim mortality findings from the trial of I-125 brachytherapy [18]. In both clinical trials, survival rates at the time of those analyses and at the end of all patient follow-up in the COMS [19,21] were essentially equivalent in the radiotherapy arm and standard enucleation arm, with neither a clinically nor statistically significant difference in mortality at any time after enrollment and treatment. Quality-of-life findings also revealed little difference between enucleation and iodine-125 brachytherapy during the first 5 years after treatment [22].

Because of the similarity of outcomes in the enucleation and brachytherapy arms, findings for the two treatment groups were pooled to provide comparisons and precise estimates of survival outcomes at 5 and 10 years within subgroups of patients and to summarize the status of patients at the end of each year of follow-up in the COMS. To our knowledge, the COMS Group published the first display of the type depicted in Figure 2. Such a visual display may be useful to physicians when counseling patients and family members regarding prognosis after primary treatment of choroidal melanoma.

The COMS brachytherapy trial resolved the issue of whether enucleation or eye-conserving radiotherapy led to earlier death or poorer quality of life than the alternative treatment for patients with medium-sized choroidal melanoma, assuming that they meet COMS eligibility criteria and are treated and followed clinically as in the COMS. Consequently, both patients who prefer to have the cancer removed from their bodies, even though removal means losing the affected eye, and those who prefer to retain their eyes for as long as possible, despite the likelihood of loss of vision, now can choose the preferred treatment without fear that they would have a longer or better life with the other treatment.

Publications

The COMS Group was fortunate that the current or former editors-in-chief of the three major ophthalmology journals were participating COMS investigators. At the times the Data and Safety Monitoring

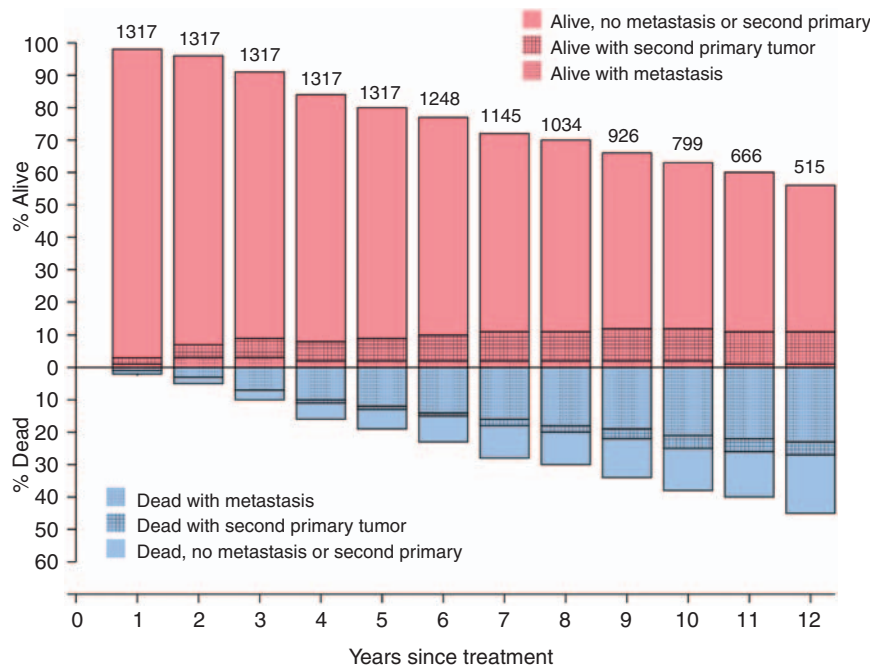


Figure 2 Percentage of patients in COMS randomized trial of I-125 brachytherapy with specified status at the end of each year of follow up (treatment arms combined)

Notes: The number of patients who enrolled early enough to be followed up to the end of the specified year is shown at the top of the corresponding bar.

Committee recommended publication of interim and final mortality findings, a decision faced by the COMS leadership was in which journal to publish the findings. The time from submission of a manuscript to publication in print was less than 6 months in each case of release of mortality findings [18–21]. In addition to mortality outcomes, the COMS investigators have published articles focused on both clinical [23–31] and methodologic [14,32–35] issues.

As noted earlier, prospective mortality follow-up of patients who were eligible for one of the COMS randomized trials but who did not enroll was considered and rejected during development and refinement of the COMS design. The investigators at one center have reported mortality findings from a subset of patients who refused or deferred treatment at the time they were evaluated and deemed eligible for the COMS [36]. A retrospective mortality study of patients eligible for one of the trials but who did not enroll was conducted by a COMS investigator with assistance from COMS personnel at several participating clinical centers [37]. Experiences in conducting these ancillary studies demonstrated the difficulties in attempting to obtain vital status information

retrospectively in the absence of centrally collected personal identifiers.

Impact

Availability and adoption of I-125 brachytherapy

A major consequence of the COMS was that radiotherapy delivered in a standard fashion became available at multiple locations, convenient for most patients diagnosed with choroidal melanoma, as a result of the standard design of the reusable gold-shielded episcleral plaques [10] and the standard brachytherapy protocol developed for the COMS. In addition to the institutions and practices that participated in the study, many institutions and practices worldwide have adopted COMS procedures for iodine-125 brachytherapy, as documented by many subsequent publications about experience with this method of treatment and by numerous inquiries and requests for the COMS brachytherapy protocol and for assistance with obtaining the materials and devices necessary for such treatment that have been received by the COMS investigators at the

Coordinating Center and at the Radiological Physics Center.

In contrast to the opposition to the COMS by the radiotherapy proponents during the planning and initiation phases, the findings from the COMS trial of brachytherapy were well received throughout the ophthalmologic and radiation oncology communities. Thus, even those opposed to the COMS randomized trial in the 1980s benefited from it by being able to provide their patients with high-quality data on which to base treatment decisions.

A possibly unfortunate outcome is that findings from the iodine-125 brachytherapy trial have been extrapolated to treatment of choroidal melanoma much larger than the tumors eligible for the COMS trial [38,39]. On the other hand, as a result of the COMS, few patients with large choroidal melanoma now undergo the additional expense and inconvenience of pre-enucleation radiation before the affected eye is removed.

Mortality rates: expected versus observed

The most significant finding from the COMS from the perspective of choroidal melanoma patients may be that absolute 5-year mortality rates among COMS patients (in both trials) were 10% lower than were estimated from published data when the COMS was designed, i.e., 20% rather than 30% in the brachytherapy trial. The 95% confidence intervals on the observed 5-year mortality rates exclude the anticipated mortality rates in both treatment arms. Long-term survival without a clinical diagnosis of metastasis or other primary cancer was better than anticipated; for example, of 515 patients followed for 12 years, 285 (55%) were alive at the end of the 12th year and 231 of the 285 (81% of those alive; 45% of all 515) were free of clinical evidence of melanoma or another primary cancer (Figure 2).

Consequences of eligibility criteria

Although COMS findings are believed to be generalizable to patients with choroidal melanoma who satisfy COMS eligibility criteria, those criteria excluded patients with clinically detectable melanoma metastasis and most other primary cancers as well as patients with other life-threatening conditions. COMS criteria also incorporated carefully defined limitations on tumor size and location. Central review of the diagnosis and tumor height and central monitoring of adherence to the radiotherapy protocols were key features of diagnosis and treatment in the COMS, but these

procedures typically are not available outside large research studies.

Patients whose tumors were contiguous with or very close to the optic disk were excluded as a consequence of the decision to adopt a radioactive plaque to deliver radiation to medium-sized tumors because of the difficulty of placing the plaque contiguous with the optic nerve. Other methods of delivering eye-conserving radiotherapy to that subgroup of patients may yield results similar to those from the COMS trial but have not been, and are unlikely to be, evaluated in randomized trials with respect to their effects on long-term survival.

Limitations and future research needs

The COMS investigators were disappointed that among patients with large choroidal melanoma, pre-enucleation radiation did not reduce mortality compared to enucleation alone, suggesting that clinically undetectable metastasis at the time of diagnosis was responsible for the deaths of most eligible patients with large choroidal melanoma as well as many patients diagnosed with medium-sized choroidal melanoma based on outcomes from the brachytherapy trial. As no randomized trial has been conducted in which any type of treatment of choroidal melanoma has been compared to no treatment, the proportion of patients in whom metastasis develops by the time choroidal melanoma is diagnosed clinically is unknown. It is possible that the similarity in findings between treatment arms in the COMS trials was solely due to metastasis present at baseline in a subset of patients enrolled. As documented by the COMS Group, survival following clinical diagnosis of metastasis is quite short with median survival of 4 months [29]. Furthermore, no effective treatment for metastatic ocular melanoma has been found to date [40]. Even treatments with some evidence of effectiveness for treating metastatic cutaneous melanoma have been found to be ineffective for treating metastatic ocular melanoma. Clearly, better methods of detecting metastases at the time of diagnosis of choroidal melanoma are needed to avoid unnecessary local treatment or to evaluate potential adjunctive treatment. Several COMS investigators are among those conducting research on detection of subclinical metastasis and treatment of metastatic ocular melanoma.

Because prevention may be more effective than treatment, epidemiologists have investigated a number of potential risk factors for choroidal melanoma with the goal of identifying those that may be modifiable. These investigations are difficult because of the rarity of the condition in the population, requiring that they be conducted at centers where a sufficiently large number of

patients with choroidal melanoma are diagnosed and treated, thus having the potential for selection bias. It is not feasible to follow large samples from the population prospectively to evaluate predictors of incident ocular melanoma. Several efforts were made to design and conduct a study of risk factors among patients with choroidal melanoma who were screened for the COMS. However, competing priorities for the time and energy of the investigators and for funding resulted in abandonment of those efforts.

Since completion of the COMS, efforts have been made by COMS and other investigators to design and conduct other randomized trials of treatments for choroidal melanoma, including a trial of adjuvant chemotherapy for large (high-risk) choroidal melanoma and a trial of early versus deferred treatment of small suspected choroidal melanoma. To date, those efforts have been unsuccessful, largely due to failure to secure the necessary financial commitment required to follow enrollees for mortality and treatment-related morbidity. Furthermore, efforts were made to obtain support to continue follow-up of COMS patients for late complications of radiotherapy as well as long-term mortality and quality of life to increase the COMS contributions to knowledge of this cancer; however, those efforts also were unsuccessful. As noted earlier, choroidal melanoma represents a very small proportion of all cancers. Furthermore, as a cause of vision loss, these ocular tumors are overshadowed by other, more prevalent ocular conditions. Thus, it is difficult to identify sponsors of research who have large enough budgets that a portion can be devoted to choroidal melanoma.

Conclusions

The COMS trial of brachytherapy versus enucleation arguably was the most challenging of the multi-center clinical trials sponsored by the National Eye Institute to organize and initiate since the Diabetic Retinopathy Study, the first multi-center randomized trial conducted under the sponsorship of that agency. Nevertheless, the multi-disciplinary collaboration required to conduct the COMS was exemplary. The primary questions addressed by the study were answered with data of good quality. More than 30 articles from the COMS Group have been published in peer-reviewed journals to report primary and secondary outcomes and to evaluate methods used in the COMS. Many ancillary studies were conducted; publication of their findings has contributed to the methodology of randomized trials and the epidemiology and treatment of choroidal melanoma. COMS investigators are

continuing to analyze and publish data collected as part of the study. COMS documents [41,42] have been deposited at the National Technical Information Service (US Department of Commerce); the COMS database together with copies of those documents and other COMS materials have been placed in the Alan Mason Chesney Medical Archives (the Johns Hopkins School of Medicine) for use by future researchers. Twenty-five years after its initiation in a climate of controversy, the COMS must be judged a success by any objective observer.

Acknowledgements

Members of the COMS Group as of different dates have been published as follows: *Controlled* 1992; 14: 387–91 (June 30, 1992); *Arch Ophthalmol* 2001; 119: 961–65 (September 30, 2000), *Am J Ophthalmol* 1998; 125: 757–65 (August 31, 1997), and *Ophthalmology* 2001; 108: 360–66 (August 31, 1999). The members of the COMS Steering Committee reviewed and approved an earlier version of this manuscript.

Funding

The COMS was supported by the National Eye Institute and, beginning in 1992, the National Cancer Institute, National Institutes of Health, US Department of Health and Human Services, Bethesda, MD, through cooperative agreements EY06253, EY06257, EY06258, EY06264, EY06265, EY06266, EY06268, EY06269, EY06270, EY06274, EY06275, EY06276, EY06279, EY06280, EY06282, EY06283, EY06284, EY06287, EY06288, EY06289, EY06291, EY06839, EY06843, EY06844, EY06848, EY06858, and EY06899.

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